Online Identification of Power System Network Branch Events

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***Abstract*—The electric power grid infrastructure is the most critically important man-made system in use today, particularly since all other infrastructures depend on the reliability of a robust electrical infrastructure. The smart grid transformation enables the real-time situational awareness through synchronized measurements. The communication network capable of bi-directional communication is responsible for the full connectivity of all the measurement and control units. In the power system, the transmission network is the bridge between the bulk generation and distribution system. The reliability of the power system is highly dependent on the successful operation of the geographically distributed transmission network and its components. Power system network branch outages, which may occur due to number of reasons, may result in either partial or complete blackout of the power system. The smart grid is a cyber-physical system. The cyber-physical systems are vulnerable to cyber and physical attacks. In this paper transmission network branch outage identification using phasor measurement units is investigated. Real-time digital simulator based simulation platform has been used. Admittance matrix based approach and neural network approach are investigated and performances evaluated.**

***Index Terms*-- Bus admittance matrix, cyber physical systems, neural networks, smart grid, transmission lines, synchrophasors**

I. Introduction

The present power system is rapidly changing towards smart grid (SG). The grid conversion from conventional system to more dynamic intelligent system is a definite requirement for sound SG operation. In a SG system, synchronized measurements and intelligent controls are used to accommodate the integration of utility scale renewables, Distributed Generation Sources (DGS), energy storage units, electrified transportation, Demand Response (DR) and demand side management. A reliable communication configuration is established to support the measurement and control equipment. The conventional Supervisory Control and Data Acquisition (SCADA) system is characterized by a low measurement report rate, approximately once every 4-6s, which is not capable of identifying low frequency electromechanical oscillations, which can lead to system blackout or brownouts. Thus, Phasor Measurement Units (PMUs) are introduced [1] in the critical network nodes. PMUs are capable of transmitting measurements at a higher rate. Typically, 30Hz reporting frequency is used in the United States. All the measurements are synchronized with Coordinated Universal Time (UTC) using Global Positioning System (GPS). All these synchronized measurements are collected into a Phasor Data Concentrator (PDC) for data management. These data represent the situational awareness of the power system, which greatly expands the possibilities for the rapid and flexible system control.

The SG is vulnerable to physical and cyber-attacks. A cyber-attack may be remote or local. Higher level of encryption tools and secure Virtual Private Network (VPN) connections are used in the communication infrastructure to protect the transmission of information. Unfortunately, such secure communications remain vulnerable to man-in-the-middle attacks such as Denial of Service (DoS) attack via the use of side channels [2]. The physical attack also can occur on any part of the power system, which can lead to catastrophic outcomes. The blackout that occurred in north eastern US in 2003 [3] is one such example of a large-scale power system failure that can occur from such an outage. Thus, implementation of protection system that react to and preempt such outages is essential for a modern and reliable power supply. Reliable monitoring system may include redundant and multiple independent verifications about system status, that can use to increase the reliability of the online system monitoring. In this paper, independent branch status identification approaches based on admittance matrix of the system and neural network are evaluated.

In Section II, the methodology of the identification approach, particularly in terms of PMU measurements extraction is detailed. In Sections III and IV the two approaches are detailed. In Section V, the results of both approaches are analysed and compared, and in Section VI the conclusions of the study are explained.

II. Methodology

An IEEE 12-bus, three area benchmark power system serves as the experimental case study, which is designed on RSCAD, a power system simulation software and simulated on the real-time digital simulator (RTDS).

This study is supported by the National Science Foundation (NSF) of the United States, under grants IIP #1312260, #1408141 and #1738902, and the Duke Energy Distinguished Professorship Endowment Fund. Any opinions, findings and conclusions or recommendations expressed in this material are those of author(s) and do not necessarily reflect the views of National Science Foundation and Duke Energy.

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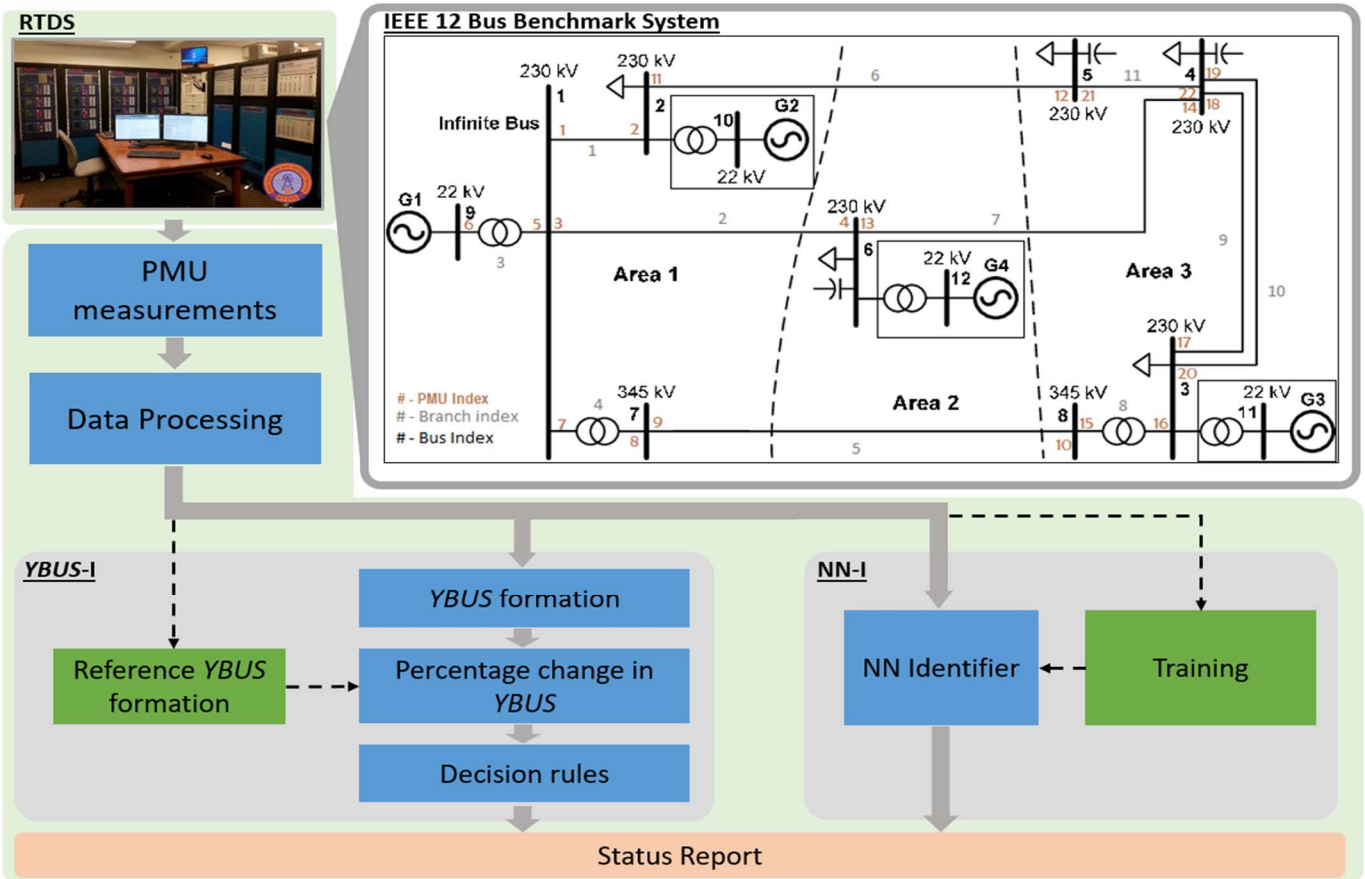

Figure 1. Power system network branch event identification tool procedure flow diagram

The IEEE 12-bus system, with three generators at bus two, three and six, has eleven branches (eight transmission lines and three transformers) in the network. The system is integrated with PMUs, measuring voltage and current of both ends of each branch. The online measurements from the PMU directly send to the computational platform for identification. The identification tool procedure for both approaches is shown in Fig. 1.

$$\text{No: of contingency events} = C_1^{11} + \cdots + C_{10}^{11} = 2046 \qquad (1)$$

The computational algorithms for both methods are implemented on MATLAB. Theoretically there are unique 2046 contingency events considering up to N-10 contingency level for this example system. The combinations calculation is shown in (1). In keeping with the standard practice, this study was conducted only for the (N-1) contingency level, thus yielding twelve operational cases (eleven single branch outages and the normal operational condition) for identification. The real-time measurements are required to transfer to the computational platform for both approaches to execute the online identification tool.

III. Admittance Matrix Based Identification (*YBUS-I*)

The power system admittance matrix (YBUS) can be used to understand system configuration. The system YBUS can form solely using the system parameters (admittance or impedance) [4]. These parameters can be derived from the component physical design parameters (e.g. for the transmission line resistance per km, inductance per km, etc.) or can be calculated using the measurements. The PMU measurements are used to calculate system YBUS frequently. YBUS is quite sparse. The study used this property of the YBUS to achieve efficient storing techniques and to use sparsity programming techniques in the identification algorithms. In the algorithms, a hybrid storage method, composed of the chained data structure storage method and the upper triangle storage method is used to maintain efficient data storage.

The transmission line equivalent π-model is shown in Fig. 2. Here, V_1, I_1, V_2 and I_2 are the voltage and current of the respective sending end and receiving end of the transmission line. Kirchhoff's Current Law (KCL) for the model shown in Fig. 2 below is then used to derive, (2) and (3).

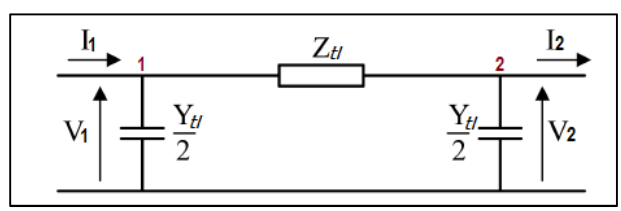

Figure 2. Equivalent π-model of a transmission line

$$I_1 = V_1\,(\mathrm{Y}_{tl}/2) + (V_1 - V_2)/Z_{tl} \qquad (2)$$

$$I_2 = V_2\,(\mathrm{Y}_{tl}/2) + (V_1 - V_2)/Z_{tl} \qquad (3)$$

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As shown in (2) and (3), $Z_{tl} = R_{tl} + jX_{tl}$ and $Y_{tl} = jy_{tl}$, using $1/Z_{tl} = g + jb$, the imaginary and the real parts of each complex quantity is separated for further matrix calculations. A single measurements frame contain all of the synchronized PMU measurements, which feed to (6) and (7). The set of eleven (7) matrices are generated for each branch to calculate the line admittance using (8).

$$[I_{tl}]_{(4\times1)} = [V_{tl}]_{(4\times3)}[Y_{tl}]_{(3\times1)} \tag{4}$$

$$[Y_{tl}] = [g\ b\ y_c]^T \tag{5}$$

$$[I_{tl}] = \left[I_1^{(Re)}\ I_1^{(Im)}\ I_2^{(Re)}\ I_2^{(Im)}\right]^T \tag{6}$$

$$[V_{tl}] = \begin{bmatrix} V_1^{(Re)} - V_2^{(Re)} & V_2^{(Im)} - V_1^{(Im)} & -\dfrac{V_1^{(Im)}}{2} \\ V_1^{(Im)} - V_2^{(Im)} & V_1^{(Re)} - V_2^{(Re)} & \dfrac{V_1^{(Re)}}{2} \\ V_1^{(Re)} - V_2^{(Re)} & V_2^{(Im)} - V_1^{(Im)} & -\dfrac{V_2^{(Im)}}{2} \\ V_1^{(Im)} - V_2^{(Im)} & V_1^{(Re)} - V_2^{(Re)} & \dfrac{V_2^{(Re)}}{2} \end{bmatrix} \tag{7}$$

(Re) and (Im) represent the real and imaginary parts of the measurement variables in (6) and (7). The simple least squares (LS) solution for $[Y_{tl}]$ of the set of linear equations expressed in (8), is used to calculate the transmission line parameters from the PMU measurements. The results are then used to form the admittances of the transmission lines.

$$[Y_{tl}] = ([V_{tl}]^T[V_{tl}])^{-1}[V_{tl}]^T[I_{tl}] \tag{8}$$

The equivalent circuit of a transformer that is reduced to a series impedance is shown in Fig. 3, assuming a negligible shunt branch. Using the same approach as above, the transformer admittance is calculated for this model using (9).

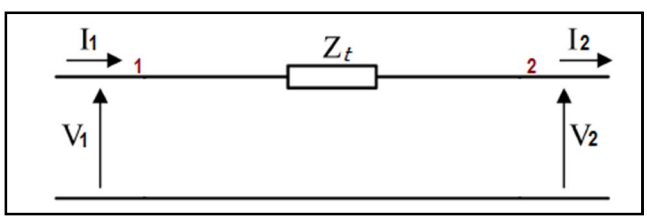

Figure 3. Equivalent circuit of a transformer

$$[Y_t] = [V_t]^{-1}[I_t] \tag{9}$$

The reference YBUS of the system refers to the power system YBUS under normal operation. The IEEE 12-bus system is simulated by tripping each branch consecutively (N-1 contingency situation) using the breaker setup in each branch of the RSCAD model. The off-diagonal elements of the YBUS of the power system represent the relevant branch configuration in between the buses that is denoted by the YBUS matrix coordinates. Diagonal elements of the YBUS proportional to accumulation of admittances of all the branches connected to the bus represent by the coordinates. The online YBUS percentage variation for a branch event that is compared with the reference YBUS is used to derive the percentage difference YBUS (PDYBUS) to enhance the decision process. The decision tree below is used to identify the branch events according to the off-diagonal and the diagonal elements of the PDYBUS. The flow diagram of the decision process in *YBUS-I* is detailed in Fig. 4.

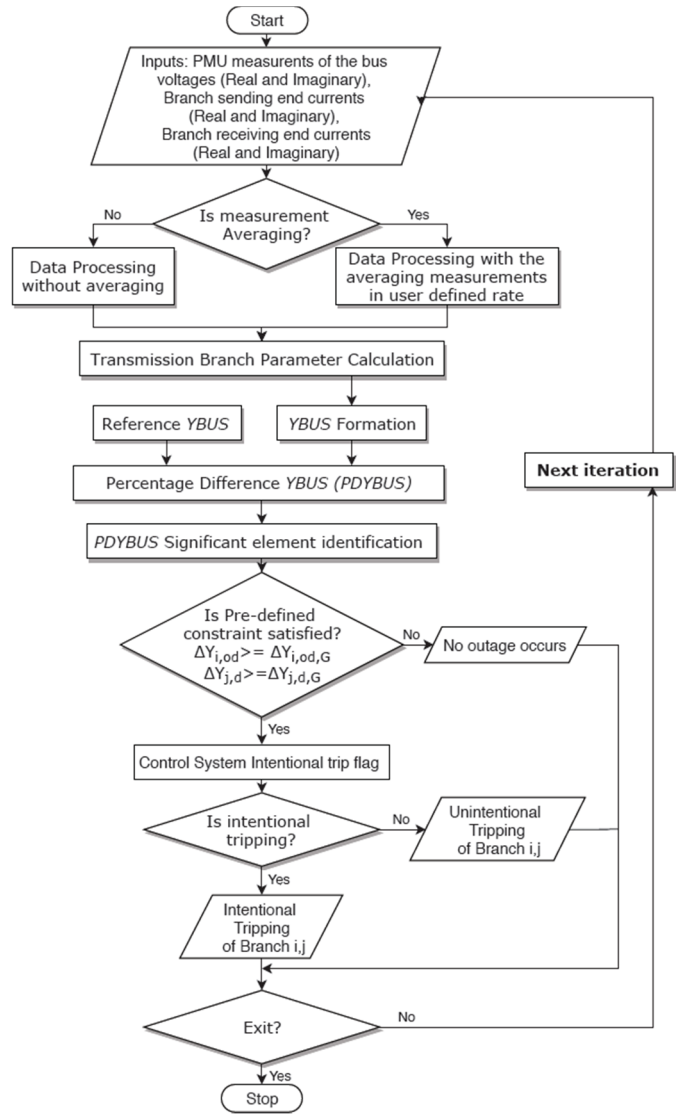

Figure 4. Flow diagram of decision tree in approach *YBUS*-I

IV. Neural Network Based Identification (*NN-I*)

The *YBUS-I* requires a higher computational power for the data arrangement, the admittance calculation, matrix calculations, the YBUS formation and decision analysis. Therefore, efficient Neural Network (NN) based learning approach is considered to reduce the identification delay [5]. The layer structure of the feed-forward multilayer perceptron NN (MLPNN), which is in Fig. 5 is used here. The MLPNN contains 62 PMU measurement inputs, ten hidden neurons and the eleven branch status as shown in Fig. 5. The system is trained with 3600 PMU measurement frames, collected from a series of random system operation cases.

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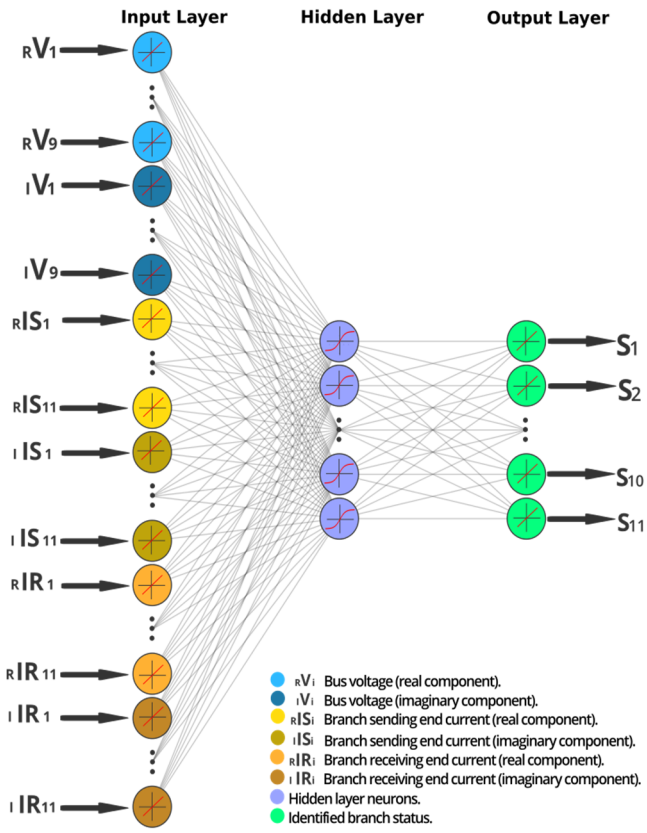

Figure 5. Layer representation of the identification in approach *NN-I*

V. Results & Discussion

The results of four discrete cases for both approaches with the actual event status are shown in Figs. 6 and 7.

1. *YBUS*-I - Not Averaged: The direct PMU measurements are used as inputs to the admittance calculations.
2. *NN-I* - Not Averaged: The direct PMU measurements are used as inputs to the MLPNN.
3. *YBUS*-I - Three Samples Averaged: The average of the three PMU measurement frames are used as inputs to the admittance calculations.
4. *NN-I* - Three Samples Averaged: The average of the three PMU measurement frames are used as inputs to the MLPNN.

Status = 1 indicates a connected line (and a closed breaker) and Status = 0 indicates disconnected line (and an open breaker). The results for both approaches without sampling are shown in Fig. 6 (i.e. the PMU measurements frame used as it is for the computation). Although the *NN-I* results show 100% identification accuracy, the *YBUS-I* has difficulty in separately identifying the serially connected branch outages. For example, the identification of events in branches 4, 5 and 8 is inaccurate in *YBUS-I* with an identification accuracy of 89.40% for case 1. The results for both approaches are shown in Fig. 7 shows with three consecutive PMU measurements averaged. In this case, the *YBUS-I* again has difficulty identifying the serially connected branch outage separation with identification accuracy of 90.73% for case 3. In both cases of *YBUS-I*, identifying parallel circuit events are not possible. As shown in Figs. 6 and 7, branch 9 and 10, both branches are considered as a single branch in *YBUS-I*. Note that Figs. 6 and 7 only shows the notable branch event graphs. The rest of the branches are correctly identified by both approaches. The comparison between two approaches is shown in TABLE I.

TABLE I. Comparisson of *YBUS-I* and *NN-I* approaches

Factors	*YBUS-I*		*NN-I*	
	Not Average	3 Samples Averaged	Not Average	3 Samples Averaged
Surety	No (<100%)		Yes (100%)	
Complexity	High		Medium	
Computational time[a]	1.12s	1.46s	0.23s	0.37s
Scalability	Low		High	
Parallel circuit separate identification	No		Yes	

a. On the same computational platform

VI. Conclusion

PMU measurements based identification of transmission network branch events was studied and demonstrated on IEEE 12-bus system. The two different branch event identification techniques used are i) admittance matrix (*YBUS-I*) and ii) neural network (*NN-I*). *YBUS-I* used the measurement to generate YBUS using KCL and least squares then form the percentage difference YBUS based on the reference YBUS run through the decision process to get the status of each branch. *NN-I* used the direct PMU measurements to establish a model using training data to identify the branch status. The comparison shows the superiority of the *NN-I* approach compared to *YBUS-I* approach. The use of PMU devices in each bus for a large power system is impractical. Therefore future research should involve i) the use of distributed approach to extended the study for a large power system and ii) investigate the identification of branch events for higher levels of contingency.

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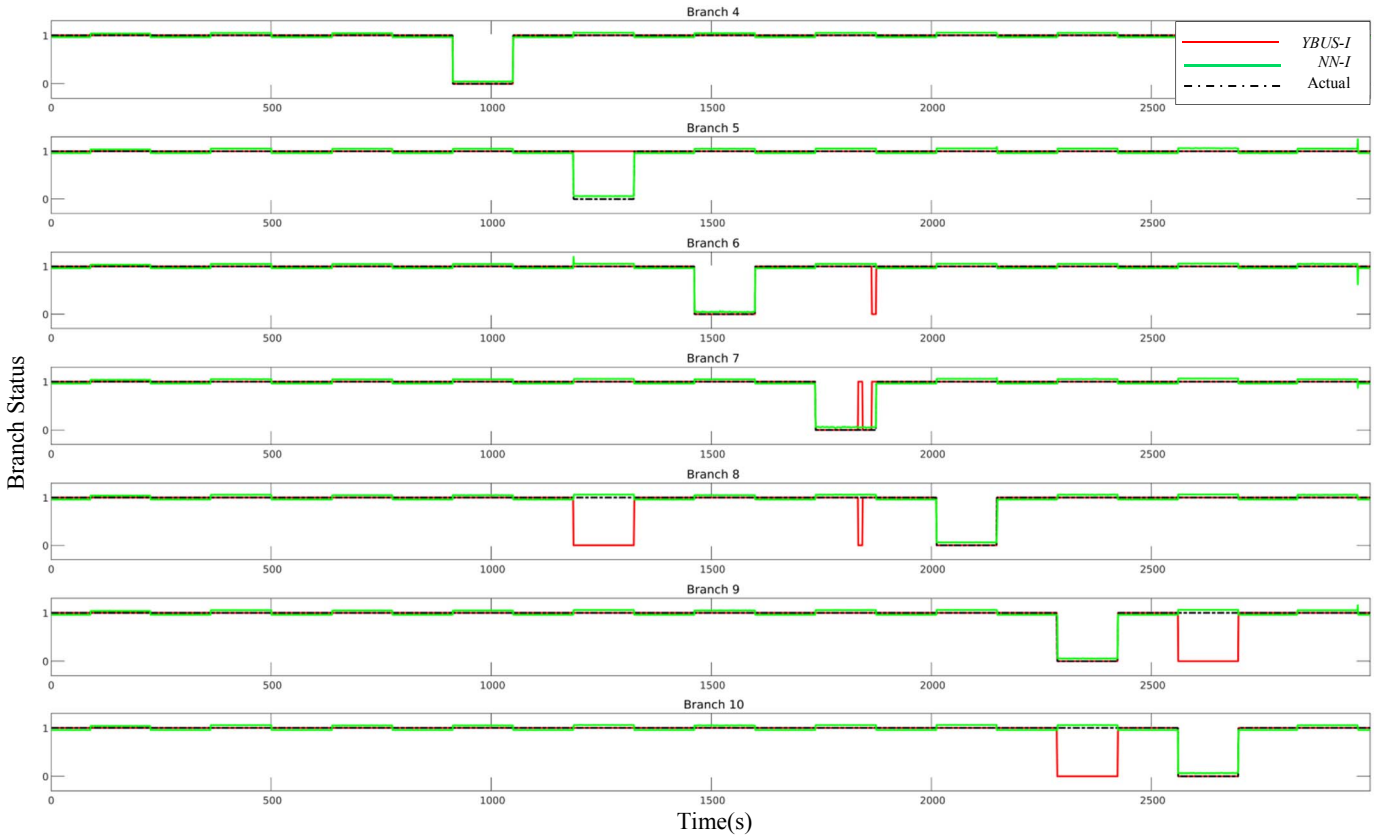

Figure 6. Graph set 1: Events of branches without measurement averaging (case 1 and 2).

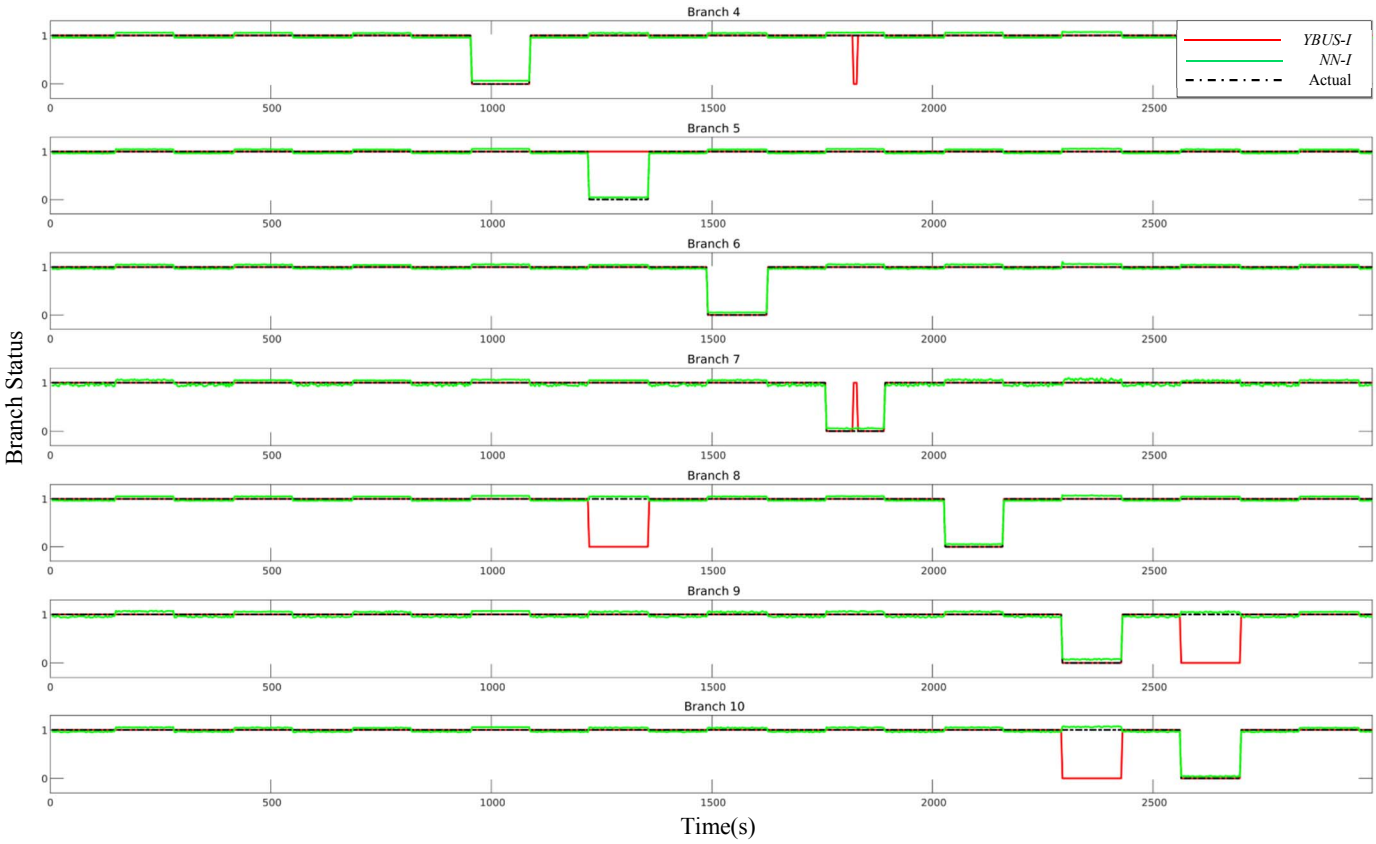

Figure 7. Graph set 2: Events of branches three measurement samples averaged as input (case 3 and 4).

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